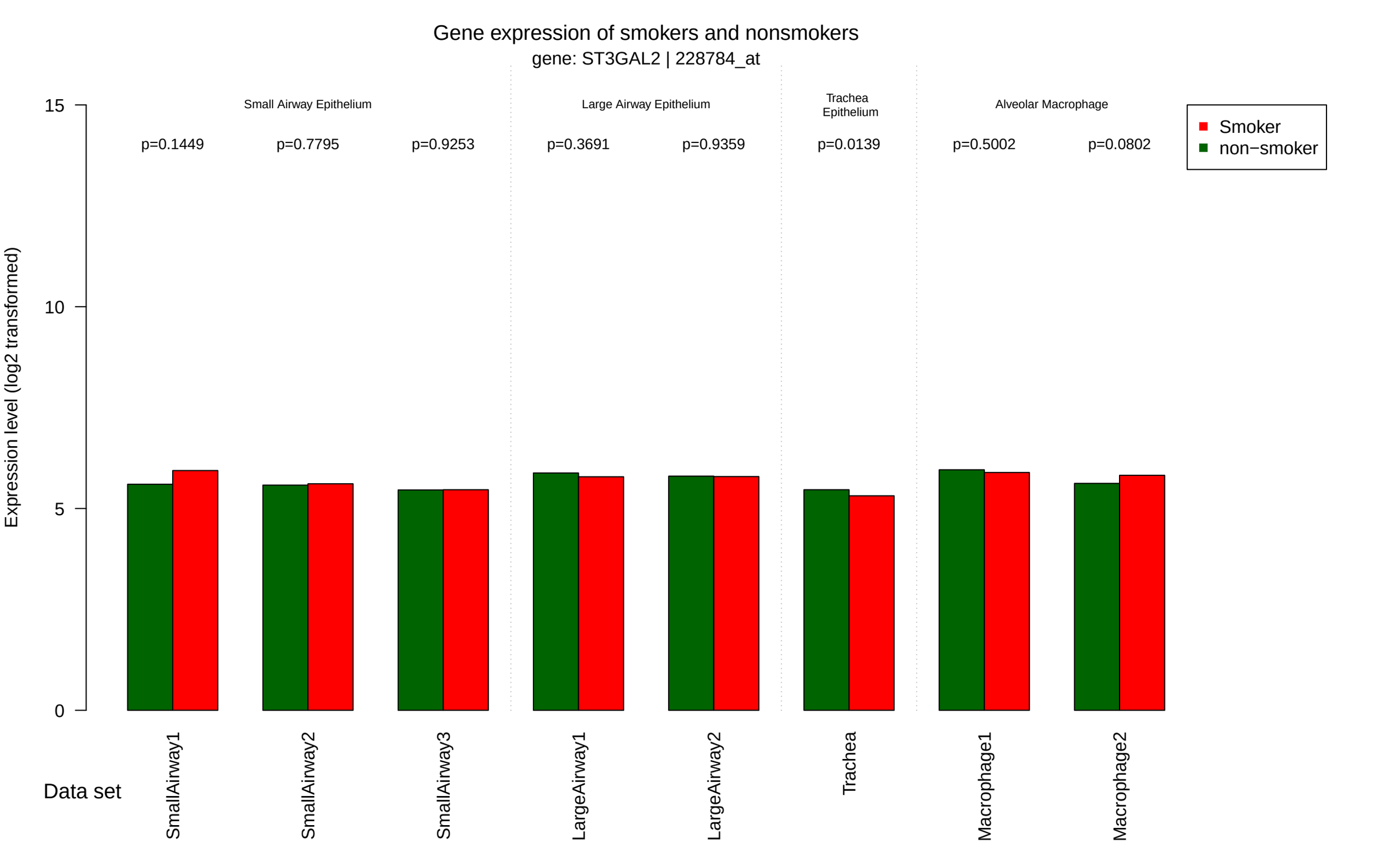
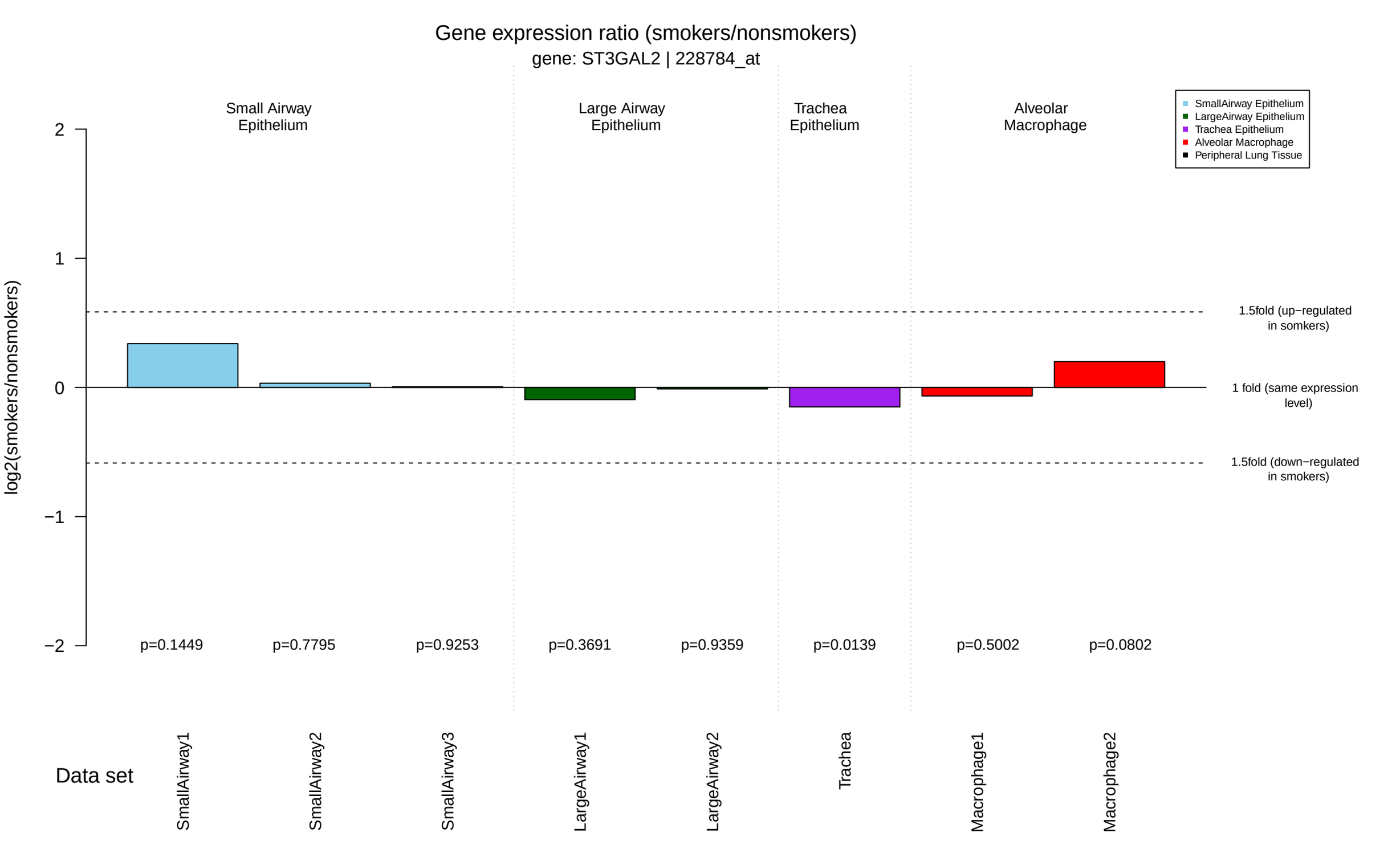
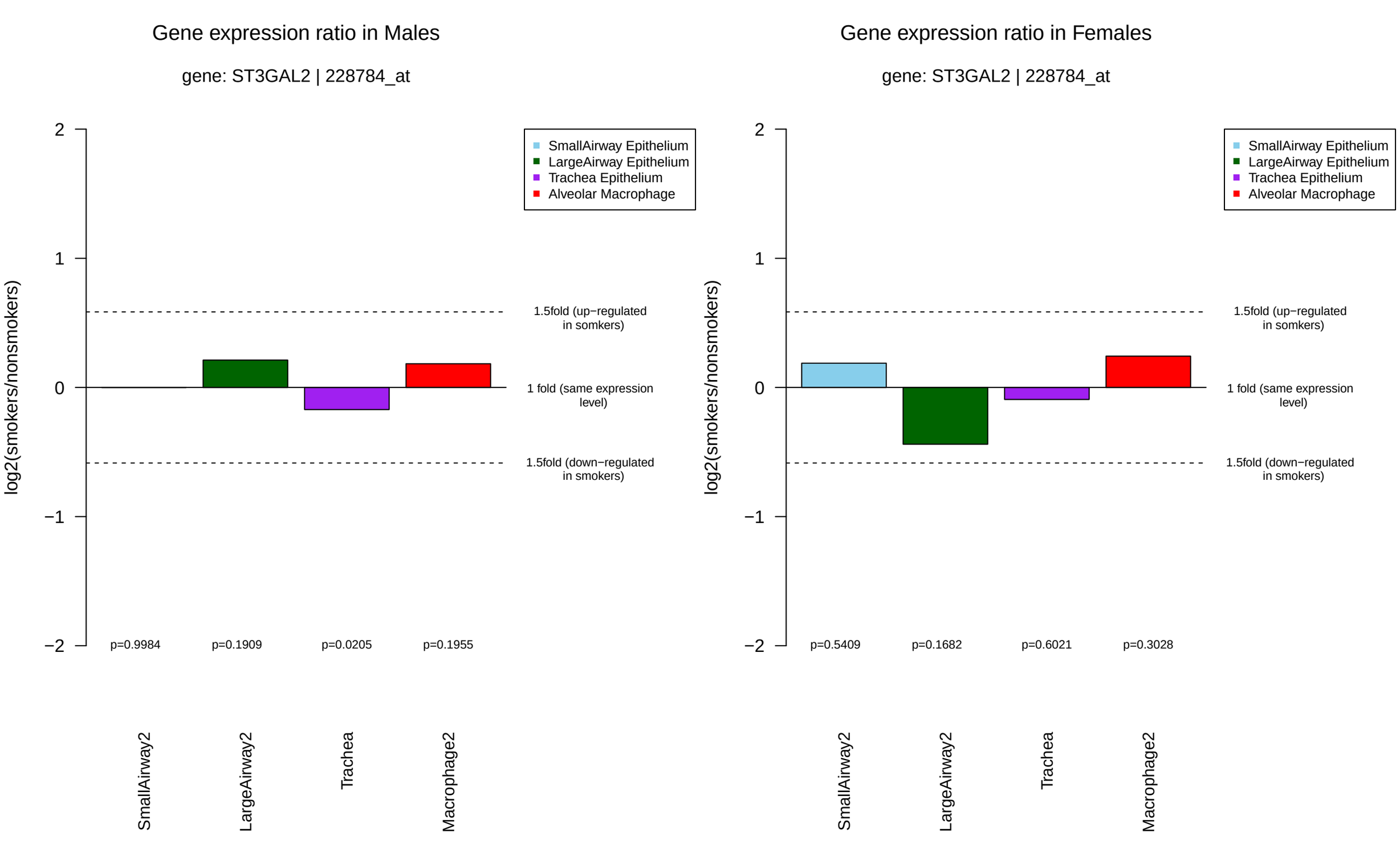
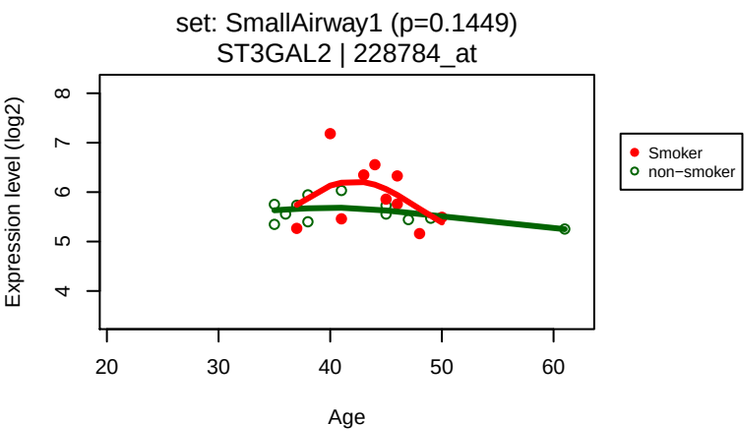
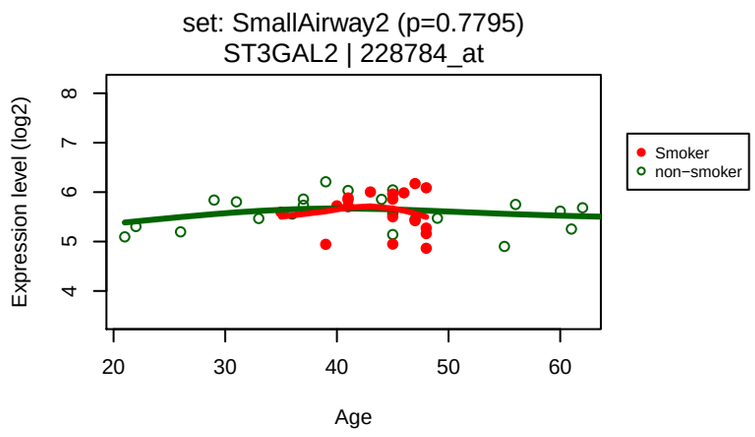
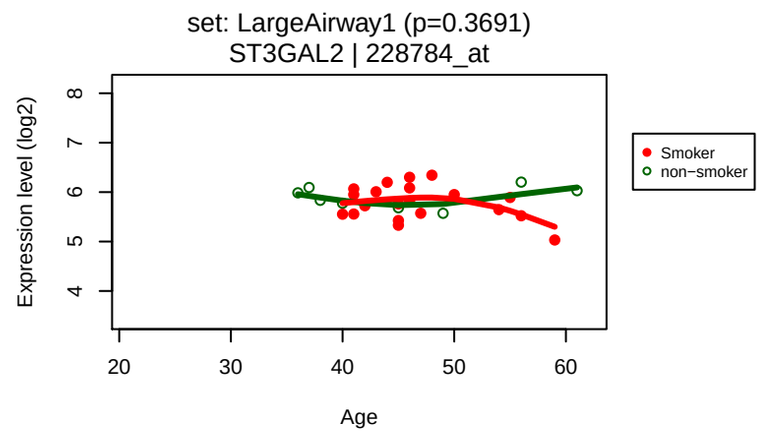

Gene expression of smokers and nonsmokers
gene: ST3GAL2 | 228784_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1449
p=0.7795
p=0.9253
p=0.3691
p=0.9359
p=0.0139
p=0.5002
p=0.0802
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

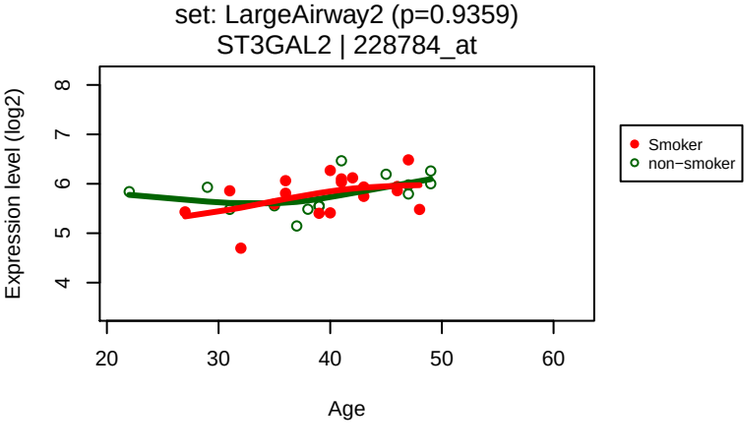

Gene expression ratio (smokers/nonsmokers)
gene: ST3GAL2 | 228784_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1449
p=0.7795
p=0.9253
p=0.3691
p=0.9359
p=0.0139
p=0.5002
p=0.0802
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

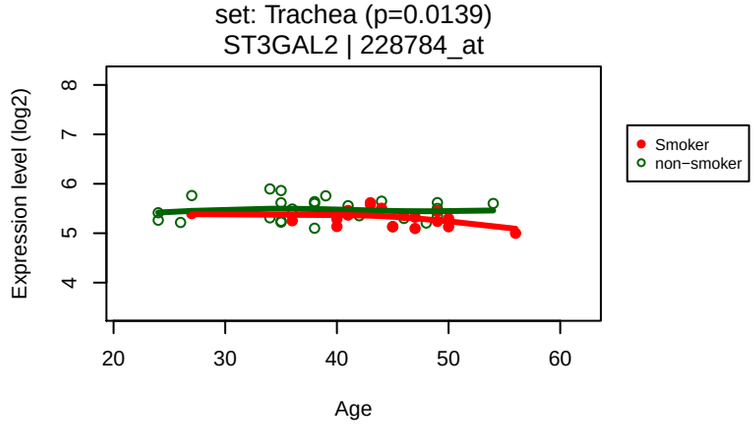

Gene expression ratio in Males
gene: ST3GAL2 | 228784_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9984
p=0.1909
p=0.0205
p=0.1955
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST3GAL2 | 228784_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5409
p=0.1682
p=0.6021
p=0.3028
SmallAirway2
LargeAirway2
Trachea
Macrophage2

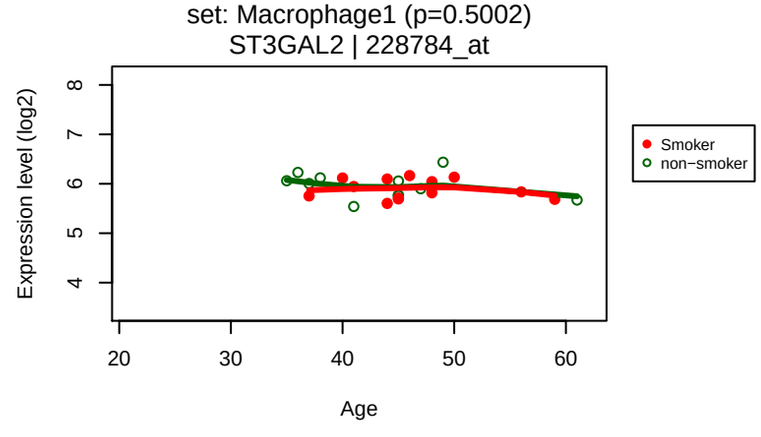

set: SmallAirway1 (p=0.1449)
ST3GAL2 | 228784_at
Expression level (log2)
Age
Smoker
non−smoker

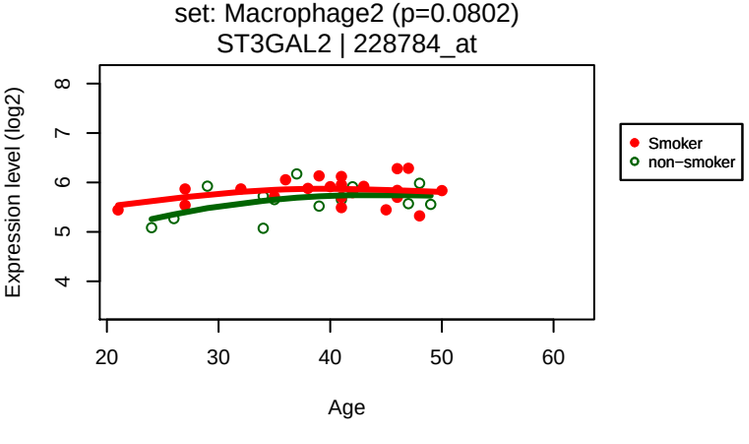

set: SmallAirway2 (p=0.7795)
ST3GAL2 | 228784_at
Expression level (log2)
Age
Smoker
non−smoker

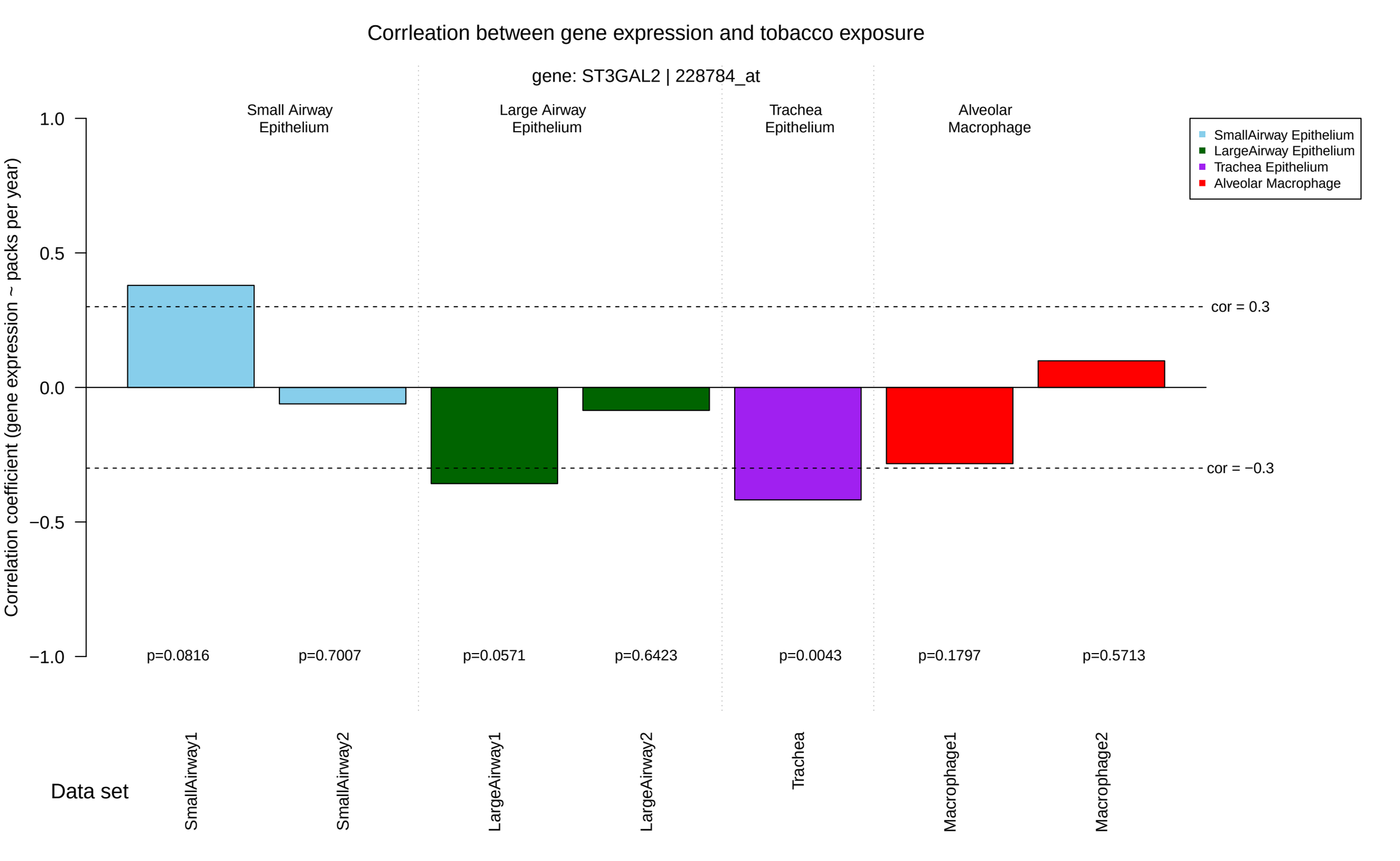

set: LargeAirway1 (p=0.3691)
ST3GAL2 | 228784_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9359)
ST3GAL2 | 228784_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0139)
ST3GAL2 | 228784_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5002)
ST3GAL2 | 228784_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0802)
ST3GAL2 | 228784_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST3GAL2 | 228784_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0816
p=0.7007
p=0.0571
p=0.6423
p=0.0043
p=0.1797
p=0.5713
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2